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SECTION 01

ASSIGNMENT 1

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In-Memory Computing in SAP HANA vs Traditional Relational Database: A Comparative Analysis for Real-Time Enterprise Analytics

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Abstract—Real-time data processing is important in modern enterprise environments to maintain a competitive advantage. Traditional relational database management systems (RDBMS), which rely on disk storage and separate transactional and analytical workloads, frequently face latency, data redundancy and delayed insights because of the complex ETL processes. These constraints are addressed by SAP HANA's in-memory computing strategy, which enables fast data access, real-time analytics and integration of OLTP and OLAP on a single platform.

This paper explores SAP HANA's architecture, key characteristics and functional principles, including hardware optimisation, memory management, stream processing and advanced analytics integration. It also looks at use cases in industries including mining and oil and gas. Improvements in performance, scalability and data processing efficiency are highlighted by a comparison with traditional RDBMS. Overall, SAP HANA is an effective solution for organisations requiring high-performance, real-time data analytics.

Keywords—SAP HANA, In-Memory Computing, Real-Time Analytics, RDBMS

I. INTRODUCTION

In today's enterprise environment, timely insights from operational data are not only beneficial, but can provide a major competitive edge. The initial purpose of traditional relational database systems (RDBMS), which rely on disk storage and row-based architecture, was to manage transactions effectively. Even if they excel at that, they frequently have trouble with analytics. Organizations usually use extensive Extraction, Transformation, and Loading (ETL) procedures to transfer data into separate data warehouses in order to fill that gap (May, Norman Lehner, Wolfgang, 2015). The disadvantage is that by the time the data is prepared for analysis, it has already become out of date, thus decision-makers may be using information that does not accurately reflect current situation.

SAP HANA overcomes this problem in a different way. It uses an in-memory design that stores active data directly in RAM rather than depending on disk-based

storage. Real-time analytics and transactional workloads are made possible by the much faster data access and processing. Businesses can now analyse real-time data without having to wait for complicated data transfers or batch updates.

This paper compares the underlying technology, performance differences and industry applications of SAP HANA against traditional disk-based database systems. By looking at these aspects, it shows how many of the limitations of traditional database structures can be addressed by modern in-memory solutions.

II. OVERVIEW OF SAP HANA

SAP HANA is a main-memory centric database system that integrates a row- and column-oriented storage engine. This architecture change enables the integration of OLTP and OLAP, allowing real-time analytics to be performed directly on real operational data without the need for data duplication. Its evolution from a purely analytical data base to an overall, comprehensive big data infrastructure that can handle huge, various workloads is highlighted by recent trends (May, Norman Lehner, Wolfgang, 2015; Kim, Kihong Kim, Hyunwook Lee, Jinsu, 2025).

III. HOW SAP HANA WORKS

SAP HANA integrates data processing, storage and advanced analytics into a single system, which makes it suitable for effectively managing heavy and complex workloads.

A. Hardware Optimization

The database system in SAP HANA is co-designed with the underlying hardware layer to take advantage of modern computer hardware such as large memory footprints, multi-core CPUs and SIMD (Single Instruction, Multiple Data) (Sikka, Vishal Färber, Franz Goel, Anil, 2013). To handle large amounts of data in a single step, it uses parallel processing and unique techniques like vectorisation.

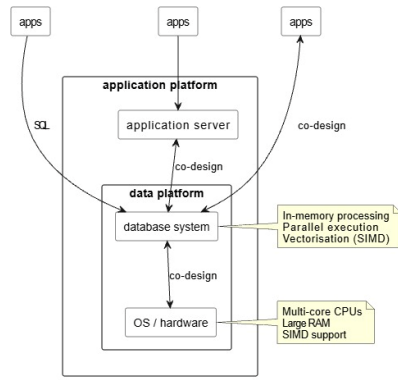


Figure 1 shows the SAP HANA platform (Sikka, Vishal Färber, Franz Goel, Anil, 2013)

As a result, it can handle large and more complex queries, especially those involving calculations and summaries, which much faster than traditional databases that depend on disk storage (Plattner, AUG 2014). In many cases, it can be more than 10 times faster, making it extremely useful for data analysis.

B. Cost-Efficient Memory Management

In-memory database systems must prioritise efficient memory management. SAP HANA solves this issue with advanced data tiering techniques, most notably the Native Storage Extension (NSE). NSE intelligently classifies data as "hot" or "warm" based on access frequency and usage patterns, ensuring that only mission-critical and frequently accessed data is stored in high-speed main memory, while less frequently used (but still relevant) data is offloaded to lower-cost disk-based storage. This automated data placement reduces pressure on expensive memory resources and can decrease overall memory use by up to 40% without causing significantly affecting query performance (Thambireddy, 2025).

Additionally, queries can still easily access both hot and warm data due to the system's transparency, even while warm data may have slightly higher latency (Thambireddy, 2025). This balance of performance and cost efficiency makes NSE extremely valuable for industries such as finance and retail, where large volumes of semi-active or historical data must be accessible for reporting, compliance and analytics without incurring excessive infrastructure expenditures.

C. Low-Latency Stream Processing

SAP HANA provides low-latency stream processing with its Streaming Analytics Server, which is designed to fulfil real-time data processing demands by reducing latency, throughput, and resource usage. Advanced performance modelling approaches improve this capability by analysing both system architecture and application logic, allowing the identification of bottlenecks and prediction of performance problems such as over-queuing situations during runtime (Rank, Johannes Paluch, Dominik, 2019).

Moreover, implementation-based studies show how SAP HANA Smart Data Streaming can effectively

use mathematical models to process continuous numerical data streams while dynamically adjusting hardware resources and system parameters to maintain optimal performance under different workloads (Nycz, 2019). Together, these methods demonstrate how well SAP HANA enables scalable and effective low-latency stream processing.

D. Advanced Analytics Integration

SAP HANA's in-memory architecture and real-time data processing capabilities are important for allowing advanced analytics integration. According to research, integrating machine learning models into SAP HANA applications, such as via custom APIs, enables effective real-time anomaly detection and improved decision-making efficiency, overcoming the constraints of built-in analytics tools (Asuah, Georgina Easin, Arafat Md Orosz, Tamas, 2025).

Furthermore, by integrating machine learning, deep learning and natural language processing (NLP) into its ecosystem alongside platforms like SAP Analytics Cloud and SAP BW/4HANA, broader AI-driven frameworks further extend the capabilities of SAP HANA and enable end-to-end analytics, including data preparation, forecasting and intelligent business insights (Jonnalagadda, Ram Reddy Reddy, Kiran Kumar Gunupati, Kamalakar, 2025). These methods demonstrate how SAP HANA serves as foundation for enterprise systems' scalable and adaptable advanced analytics integration.

E. Inter-Process Communication (IPC)

Inter-Process Communication (IPC) affects how well in-memory data analytics systems perform, especially when large amounts of data move between different processing stages. In high-performance environments, traditional IPC techniques like pipes and sockets sometimes cause latency and turn into bottlenecks. By integrating data processing directly into the database management system, SAP HANA may reduce its reliance on IPC and prevent costly data transfers between different processes (Meng, Qing-Zhong Zhou, Xuan Wang, Shan, 2018). However, this closely linked design may not be suitable in all applications, leading to alternate solutions such as advanced shared-memory techniques that allow for efficient data sharing across loosely coupled processes. These improvements emphasise how important it is to optimise IPC methods in order to fully utilise the benefits of in-memory computing in SAP HANA and related platforms.

IV. REAL-WORLD APPLICATIONS CASE STUDY

SAP HANA is a powerful platform used across many sectors to enable real-time data processing, advanced analytics and high-performance computing. The following case studies highlight its practical applications.

A. Mining Industry: Real-Time Data Integration

XYZ Corp, an Indonesian gold mining company, integrated SAP Business Warehouse with SAP

S/4HANA, which is supported by SAP HANA. Before this implementation, the company used outdated systems that needed to be manually updated every month, which caused delays in the availability of data and ineffective decision-making.

With SAP HANA's in-memory computing capabilities, company was able to integrate data in real time across multiple mining locations. This enabled faster data processing, automated data modelling and advanced analytics. As a result, enterprise-wide visibility was improved and operational efficiency was increased (Saputra, Muhardi Saputra, Rangga Ardi, 2023).

B. Oil and Gas: Predictive Maintenance and Intelligent Operations

In the oil and gas company, SAP HANA supports SAP S/4HANA Rise Cloud, allowing for real-time data processing in cloud environments. SAP HANA enables predictive maintenance and operational optimisation easier when integrated with cloud computing and artificial intelligence.

By processing a huge quantity of sensor and operational data in real time, the platform enables companies to predict equipment failures, lower maintenance costs and enhance safety standards. SAP HANA also helps sustainability by optimising resource utilisation and improves decision-making with real-time analytics. These capabilities show its significant impact on efficiency and innovation in the oil and gas sector (Nendrambaka, 2025).

	Färber, Franz Goel, Anil, 2013)	
Data Tiering	Native Storage Extension (NSE), hybrid hot/warm data management (Thambireddy, 2025)	Limited tiering, mostly disk-based (May, Norman Lehner, Wolfgang, 2015)
Application Design	SQLScript, in-database analytics, paradigm shift required (Sikka, Vishal Färber, Franz Goel, Anil, 2013)	Application logic outside DB, standard SQL (Faerber, 2011)
Scalability	Distributed scale-out, ACID, real-time analytics at scale (Plattner, AUG 2014; Nendrambaka, 2025)	Scale-up, limited distributed capabilities (May, Norman Lehner, Wolfgang, 2015)
Industry Use Cases	Finance, retail, manufacturing, life sciences, healthcare, big data (Asuah, Georgina Easin, Arafat Md Orosz, Tamas, 2025; Saputra, Muhardi Saputra, Rangga Ardi, 2023)	General enterprise, less suited for real-time analytics (Nycz, 2019)

V. COMPARATIVE ANALYSIS: SAP HANA VS TRADITIONAL RDBMS

TABLE I COMPARISON OF SAP HANA AND TRADITIONAL RDBMS

Aspect	SAP HANA (In-Memory)	Traditional RDBMS (Disk-Based)
Architecture	Main-memory centric, hybrid row/column store, OLTP/OLAP integration (Sikka, Vishal Färber, Franz Goel, Anil, 2013)	Disk-based, row-oriented, OLTP/OLAP separation (Faerber, 2011)
Performance	SIMD, multi-core optimized, low-latency, high throughput (Sikka, Vishal Färber, Franz Goel, Anil, 2013)	Limited hardware optimization, higher latency (Plattner, AUG 2014)
Mixed Workloads	Efficient OLTP/OLAP integration, workload management, resource scheduling (Sikka, Vishal	Separate systems, ETL overhead, data freshness issues (May, Norman Lehner, Wolfgang, 2015)

VI. CONCLUSION

SAP HANA represents a major step forward in database technology, migrating from traditional disk-based systems to an in-memory, hardware-optimized platform. It reduces transmission of data delays and eliminates the need for separate systems by integrating transactional and analytical processes.

This study explains how SAP HANA can provide better performance and real-time insights than traditional RDBMS because of features such as parallel processing, vectorisation, efficient memory management and advanced analytics integration. The comparative analysis also emphasises its capabilities for handling mixed workloads, scalability and support for modern enterprise applications.

Case studies show that SAP HANA enhances operational efficiency, improves decision-making and supports innovation across industries. However, infrastructural investment and application design modifications might be required for adoption.

Overall, SAP HANA is a powerful and scalable solution for organizations requiring real-time analytics and high-performance data processing.

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Logbook

Date	Search Keywords	AI Prompts Used	Task Description
12 April 2026	SAP HANA overview, in-memory database	None	Researched papers on SAP HANA by Scopus and Web of Science.
15 April 2026	SAP HANA architecture, system design, SIMD, vectorization	What is vectorization?	Wrote introduction, overview of SAP, system architecture and its section hardware optimization and cost-efficient memory management
19 April 2026	Stream processing, real-time analytics, data processing performance	What is IPC in SAP HANA?	Completed sections of low-latency stream processing, advanced analytics integration, inter-process communication (IPC)
21 April 2026	Case study, retail, manufacturing, big data	None	Researched and analyzed real world applications
25 April 2026	SAP HANA vs RDBMS comparison	Compare SAP HANA and traditional databases	Analyzed and completed comparative analysis section
26 April 2026	APA format reference	None	Finalized the report by completing the abstract and conclusion, and reviewing overall content